ViralSafeTarget: evidence-aware, virus-first CRISPR target prioritization with an exhaustive HSV-2 case study

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Abstract

Background

Guide-design software commonly emphasizes sequence-level activity features and predicted host off-targets. Viral target selection poses a broader decision problem: candidate sites should also be evaluated across viral strains, mapped to gene and protein context, distinguished from gene-level targetability, linked to biological evidence with species provenance, and considered as multiplex panels with explicit sequence-level escape assumptions.

Results

We developed ViralSafeTarget, an open and reproducible framework that keeps these decision axes separate. In an exhaustive herpes simplex virus 2 (HSV-2) case study, 28,578 initial candidate coordinates yielded 23,108 eligible rows and 21,654 unique guide sequences. A completed 109-batch Cas-OFFinder search against GRCh38.p14 retained 440,341 predicted human-match rows; 2,668 candidate-coordinate rows had no predicted human hit within the declared assembly, PAM and mismatch model. UL3, UL10 and UL52 led the gene-targetability ranking, whereas the highest-ranked individual guide mapped to UL36, whose gene-level portfolio ranked 21st. A frozen 257-guide panel produced 271 guide-to-CDS mappings, 5,691 bounded indel hypotheses and 17,733 single-nucleotide exact-target counterfactuals. Four configured three-guide panels each required at least three distinct substitutions to remove every exact guide/PAM target; this is a sequence-level barrier, not an evolutionary probability. On the same frozen panel, Cas-OFFinder and CRISPRitz host-search ranks showed Spearman correlation 0.880 and top-50 overlap 49/50. ViralSafeTarget pre- and post-host ranks correlated at 0.962.

Conclusions

ViralSafeTarget provides auditable decision support from viral genomes to research shortlists while preserving missing data and evidence provenance. The HSV-2 analysis shows that top individual guides, targetable genes and biologically established targets are distinct concepts. Results are computational hypotheses for independent review and do not establish editing, host safety, viral inhibition, delivery, treatment or cure.

Keywords: CRISPR; viral genomics; HSV-2; guide RNA; off-target screening; multi-strain conservation; evidence provenance; reproducible bioinformatics

Background

CRISPR-associated nucleases are programmable by short guide sequences, making target selection a central computational step in genome editing [1]. Mature tools such as CRISPOR and CHOPCHOP support guide design, activity scoring and predicted off-target assessment [2,3], while Cas-OFFinder, CRISPRitz and GuideScan2 emphasize scalable specificity or off-target analyses [4–6]. These capabilities are complementary rather than interchangeable.

For a virus, a technically attractive guide may be absent from circulating strains, may fall in a gene of uncertain relevance, or may produce limited predicted coding disruption. Conversely, a classically important viral gene may offer fewer conserved and host-distinct sites. A single composite “therapeutic” score would obscure these differences and overstate what sequence analysis can establish.

ViralSafeTarget was designed as a virus-first framework that integrates existing alignment and host-search engines with reproducible viral-population analysis, one-to-many annotation mapping, separate guide- and gene-level rankings, bounded coding-disruption hypotheses, source-provenanced literature evidence, and configured multiplex comparisons. Its intended output is an auditable research shortlist, not a clinical recommendation.

We report the software architecture and an exhaustive HSV-2 computational case study. The case study tests three methodological propositions: (i) guide rank and gene-level targetability can diverge; (ii) model-bounded host-search agreement is an axis distinct from virus-first composite ranking; and (iii) exact-target robustness of a multiplex panel can be represented reproducibly without claiming evolutionary probability.

Implementation

Software architecture and project contract

ViralSafeTarget 0.10.0 is implemented in Python and distributed as the `viral-safe-target` package with the `vst` command-line interface. A project is defined by versioned virus, host and nuclease profiles plus a project configuration. The same public commands support initialization, validation, planning, execution, resumption, status reporting, opening a primary HTML report, and exporting a portable result bundle.

Each stage records input checksums, parameters, tool versions, explicit missing or external-required states, measured timings and output paths. Cached stages are reused only when their signatures remain valid. Missing external results are never converted to zero predicted hits.

Viral inputs, quality control and conservation

The HSV-2 reference was RefSeq accession NC_001798.2. Discovery used a frozen set of 14 accepted HSV-2 genomes after explicit quality-control rejection of incomplete, length-incompatible or ambiguity-heavy records. Viral sequences were aligned with MAFFT [7]. Candidate support was measured as exact protospacer-plus-PAM presence across accepted genomes; a separate held-out sequence collection was retained for population support analysis when loci were observable.

Candidate enumeration and annotation mapping

The reference genome was scanned for configured SpCas9 targets consisting of a 20-nucleotide protospacer adjacent to an NGG PAM. Sequence features included GC balance, complexity, viral occurrence counts, exact strain coverage and annotation context. Coordinates were mapped to GFF3 gene and CDS features. Overlapping annotations were preserved as one-to-many mappings rather than collapsed to a single label. CDS strand and phase were used to map cut coordinates to coding and protein positions.

Host-genome screening

Eligible candidate sequences were screened against the GRCh38.p14 reference assembly (GCF_000001405.40) with Cas-OFFinder 2.4.1 [4], using the configured NGG PAM and up to three mismatches. The exhaustive workflow was divided into 109 checksum-resumable batches. A completed search with zero returned sites was distinguished from a pending, failed or missing batch. “Zero predicted hit” therefore always refers to this bounded reference-assembly search and is not evidence of biological safety.

Guide and gene targetability

Guide-level ranks retained conservation, viral uniqueness, GC, sequence complexity, annotation and model-bounded host-search components with transparent explanations. Gene-level targetability was computed separately from the portfolio of eligible and screened candidates per annotated gene. It included the best candidate, robustness of leading candidates, the lower confidence bound on the clean fraction, host-hit burden and conservation. Biological evidence was reported separately and was not added to the exhaustive gene-targetability score.

Virtual coding-disruption hypotheses

For each guide-to-CDS mapping, integer indel sizes from −10 through +10 bp were enumerated. Indel class, reading-frame remainder, determinable premature-stop position, retained protein fraction and overlap with optional domain or disorder annotations were reported. Insertions were represented by size only when their sequence was unknown. Fractions over the equally weighted size grid are descriptive hypotheses and are not repair probabilities.

Exact-target escape counterfactuals and multiplex panels

All three alternative bases at every protospacer and PAM position were enumerated for the frozen panel. A substitution was classified by whether it removed the exact protospacer/PAM target under the configured IUPAC PAM pattern. For each configured multiplex panel, an exact set-cover calculation identified the minimum number of distinct substitutions required to remove all exact sites. This quantity omits mutation rates, mismatch-tolerant cutting, viral fitness, selection, linkage and population dynamics; it is not an evolutionary escape probability.

Evidence Agent and mandatory human review

The Evidence Agent generates gene- and virus-aware queries, retrieves structured records from PubMed, Europe PMC, UniProt and NCBI, and proposes evidence rows with tested virus, experiment, phenotype, source and directness. AI-generated proposals do not affect biological scores until approved by a named domain-qualified reviewer. Evidence from HSV-1 orthologs remains separate from direct HSV-2 evidence. Biological interpretations marked with an asterisk in this working manuscript remain pending human verification.*

Systematic multi-tool benchmark

A 257-guide panel was frozen before comparison. ViralSafeTarget pre-host and post-host ranks, Cas-OFFinder hit-burden ranks, and CRISPRitz 2.6.6 profile ranks were normalized within tool. The independent CRISPRitz run used its official container image digest, GRCh38.p14, NGG, up to three mismatches, eight threads, and no bulge or population-variant options [5]. Raw scores from different systems were not averaged. CRISPOR, CHOPCHOP and GuideScan2 were included in a source-based capability matrix but remained export-required in the quantitative benchmark because no raw export was committed.

Rank agreement used Spearman correlation on shared candidates. Top-k overlap was reported for k = 10, 25 and 50. A leave-one-component-out analysis recomputed the transparent ViralSafeTarget score after omitting each configured component; it is a sensitivity diagnostic, not model training or biological validation.

Reproducibility, portability and generative-AI disclosure

The installed wheel was tested outside the repository checkout. A reference-only BK polyomavirus workflow (NC_001538.1) used the same CLI and schemas without a virus-specific branch in core code. Unavailable host and population stages remained explicitly unavailable. This demonstrates software portability, not biological generalization.

Generative AI assisted with code organization, documentation, literature-query preparation and manuscript language. All numerical outputs reported here were generated by versioned software from committed machine-readable sources and are checked by build assertions. AI is not an author. Biological claims and source interpretations marked with an asterisk remain pending author or domain-expert verification.*

Results

Exhaustive HSV-2 source validation

Table 1. Exhaustive HSV-2 analysis summary

Metric	Verified value
Initial candidate coordinates	28,578
Eligible candidate-coordinate rows	23,108
Unique guide sequences	21,654
Cas-OFFinder batches completed	109/109
Predicted human-match rows	440,341
Zero-hit candidate rows under model	2,668

Values are computational and model-bounded; missing evidence remains unknown.

The publication build revalidated every primary count before generating tables or figures (Table 1; Fig. 2). From 28,578 initial candidate coordinates, 23,108 were eligible after sequence and annotation filters. These corresponded to 21,654 unique guide sequences. All 109 Cas-OFFinder batches completed. The committed host-search table contained 440,341 predicted human-match rows, and 2,668 candidate-coordinate rows had no predicted match under the configured model.

Guide rank and gene targetability diverged

Table 2. Selected HSV-2 gene-targetability results

Gene	Rank	Score	Zero-hit rows	Best candidate
UL3	1	0.882	24	VST-5e4034e33559c37a
UL10	2	0.805	47	VST-2b3568cdc187cc3a
UL52	3	0.796	89	VST-0bf1ec26dd4e63a9
UL47	4	0.774	102	VST-9cd67af4bd66d50a
UL11	5	0.723	14	VST-b3c226e1dbf82116
UL30	8	0.706	73	VST-240e20eb666f9c85
UL18	12	0.671	29	VST-7ca830d5c47bd4fc
UL19	20	0.617	110	VST-2ef5158024ade344
UL36	21	0.605	159	VST-2e9f052157f9bf29

Values are computational and model-bounded; missing evidence remains unknown.

UL3, UL10, UL52, UL47 and UL11 occupied the first five exhaustive gene-targetability ranks (Table 2; Fig. 3). The highest-ranked individual candidate, VST-2e9f052157f9bf29, mapped to UL36. UL36 ranked 21st at gene level, whereas UL3 led the gene portfolio despite not containing the top individual guide. Thus the best single candidate and the strongest gene-level portfolio were empirically distinct.

This result does not imply that UL3 is biologically or therapeutically preferable to UL30 or UL52. Direct HSV-2 essentiality evidence for the focus genes remains incomplete.* HSV-1 functional studies provide replication-focused context for UL30 and UL52, but those observations remain ortholog evidence rather than direct HSV-2 null evidence.*

The deep panel added coding and sequence-robustness context

Table 3. Frozen-panel virtual analysis

Metric	Value
Frozen guides	257
Guide-to-CDS mappings	271
Guides mapped to at least one CDS	250
Bounded indel hypotheses	5,691
Single-nucleotide counterfactuals	17,733
Held-out observations available	200/257

Values are computational and model-bounded; missing evidence remains unknown.

The frozen deep panel contained 257 unique guides. Generic mapping produced 271 guide-to-CDS rows because overlapping CDS records were retained; 250 guides mapped to at least one CDS (Fig. 4). The −10 to +10 bp grid generated 5,691 bounded indel hypotheses. The analysis enumerated 17,733 single-nucleotide protospacer/PAM counterfactuals. Held-out exact-target coverage was available for 200 guides and remained unknown for 57.

For the configured focus genes, all three analyzed UL18 guides intersected the supplied domain annotation and occurred relatively early in the protein, giving UL18 a strong predicted-disruption context.* This is annotation-dependent and does not demonstrate an HSV-2 phenotype. UL36 contributed seven deep-panel guides, including the leading individual candidate, but its gene-level portfolio remained lower ranked.

Multiplex panels had a three-substitution exact-target barrier

Four three-guide strategies were evaluated: top-ranking-only, essential/replication-focused, targetability-focused and mechanism-diverse (Fig. 5). Each required three distinct substitutions to remove all exact guide/PAM sites under the set-cover model. Minimum available held-out support ranged from 0.973913 to 0.982609. These are marginal per-guide values and do not establish within-genome joint coverage or reduced biological escape.

Host-search engines agreed while composite ranks addressed a different axis

Table 4. Frozen-panel benchmark summary

Comparison	Result
Cas-OFFinder vs CRISPRitz Spearman	0.880413
Top-10 overlap	9/10
Top-25 overlap	24/25
Top-50 overlap	49/50
VST pre-host vs post-host Spearman	0.961975

Values are computational and model-bounded; missing evidence remains unknown.

Cas-OFFinder and CRISPRitz ranks correlated at 0.880413 over all 257 frozen guides (Fig. 6). Their top-k overlaps were 9/10, 24/25 and 49/50. ViralSafeTarget pre-host and post-host ranks correlated at 0.961975 and retained identical membership at top 10, 25 and 50. In contrast, host-burden ranks showed low correlation with ViralSafeTarget composite ranks (0.085008–0.296323), as expected because the latter also incorporated viral conservation and annotation context.

The leave-one-component-out analysis showed greatest sensitivity to sequence complexity within this enriched panel: omitting it caused median absolute rank shift 75, maximum shift 230 and top-10 overlap 0 (Supplementary Fig. S1). Conservation, viral uniqueness, annotation and gene-evidence columns were constant in the selected panel, so their omission produced no rank change. This diagnoses the frozen panel and scoring configuration, not predictive correctness.

The installed workflow generalized operationally to a second virus

The 0.10.0 wheel completed initialization, planning, execution, resumption, reporting and export outside the source checkout. The same installed interface retrieved and processed BK polyomavirus NC_001538.1 without core-code modification. It estimated 543 editor-compatible sites before a configured 500-row cap and completed annotation, virtual analyses, multiplex comparison and export. Host screening and population conservation remained unavailable, as no host assembly or multi-strain panel was supplied.

Discussion

ViralSafeTarget addresses a decision problem that extends beyond guide generation. Its principal contribution is the auditable integration—and deliberate separation—of viral population support, host-search status, guide rank, gene-level targetability, predicted coding disruption, evidence provenance and exact-target multiplex robustness.

The HSV-2 case study illustrates why these axes should not be collapsed. UL36 contained the leading individual guide but provided a weaker gene portfolio. UL3 led gene targetability, while its direct HSV-2 biological importance remains unresolved.* UL30 and UL52 retained clearer HSV-1 replication-focused rationales*, yet did not occupy the first two targetability ranks. These differences define testable research questions; they do not establish therapeutic priority.

The independent host-search comparison increases confidence that the reported reference-genome burden is not unique to one search engine under matched settings. However, agreement cannot establish safety, and divergence from the composite rank is not evidence that ViralSafeTarget is more accurate. The systems answer overlapping but different questions.

To our knowledge, we did not identify an existing platform that jointly integrates multi-strain viral conservation, model-bounded host-genome risk, separate guide- and gene-level targetability, source-provenanced biological evidence, bounded coding disruption, and configured exact-target multiplex escape analysis in one reproducible project workflow. Individual capabilities are available in established tools, and the claim concerns integration rather than invention of each component.

The intended use is early-stage computational prioritization and transparent handoff to domain experts. Future work should complete human evidence review, obtain quantitative exports from additional guide-design systems, evaluate a fully multi-strain second virus, recruit external users, and compare computational ranks with experimental ground truth.

Conclusions

ViralSafeTarget converts viral reference, population, annotation and host inputs into an auditable research shortlist with explicit uncertainty and provenance. Its HSV-2 case study demonstrates that guide quality, gene targetability, host-search burden, predicted disruption, evidence coverage and exact-target robustness are distinct properties. The software and frozen computational outputs support independent review and reproducibility, but not claims of editing, safety, viral inhibition, treatment or cure.

Availability and requirements

- **Project name:** ViralSafeTarget
- **Project home page:** https://github.com/ronnuriel/ViralSafeTarget
- **Archived version:** [DOI PENDING]
- **Operating systems:** Platform-independent Python; external engines may have platform-specific requirements
- **Programming language:** Python 3.10 or later
- **Other requirements:** Optional MAFFT, Cas-OFFinder and CRISPRitz for corresponding stages; missing tools remain explicit
- **License:** MIT
- **Restrictions for non-academic use:** None under the MIT license

List of abbreviations

CDS: coding sequence; CI: confidence interval; CRISPR: clustered regularly interspaced short palindromic repeats; GFF3: General Feature Format version 3; HSV-1: herpes simplex virus 1; HSV-2: herpes simplex virus 2; PAM: protospacer-adjacent motif; SNV: single-nucleotide variant; VST: ViralSafeTarget.

Declarations

Ethics approval and consent to participate

Not applicable. This computational study used publicly available viral and reference genome data and did not involve human participants, human tissue or animals.

Consent for publication

Not applicable.

Availability of data and materials

Source code and compact result snapshots are available at https://github.com/ronnuriel/ViralSafeTarget. The archived software and supporting data release will be cited here after DOI minting: [DOI PENDING]. Public sequence accessions, configuration files, checksums and provenance manifests are included in the repository and additional files.

Competing interests

The author declares no competing interests. [AUTHOR CONFIRMATION REQUIRED]

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Authors' contributions

RN conceived the project, implemented and evaluated the software, interpreted the computational findings and prepared the manuscript. [AUTHOR CONFIRMATION REQUIRED]

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Generative AI systems assisted with code organization, documentation, query preparation and language editing under author supervision. AI systems are not authors. Biological claims marked with an asterisk remain pending human verification. [AUTHOR CONFIRMATION REQUIRED]

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Figure legends

Figure 1. ViralSafeTarget separates complementary viral target decision layers. Versioned inputs pass through viral conservation, annotation, host screening, separate guide and gene ranking, coding-disruption hypotheses, exact-target robustness and human-reviewed evidence. Outputs remain research candidates rather than treatment recommendations.

Figure 2. Exhaustive HSV-2 screen and bounded host-search funnel. Counts are verified from committed source tables. The 2,668 zero-hit rows had no predicted GRCh38.p14 match through three mismatches under the configured SpCas9 model; this is not evidence of safety.

Figure 3. Individual-guide rank diverges from gene targetability rank. Gene targetability ranks use candidate portfolios. The leading individual guide maps to UL36, while UL36 ranks 21st as a gene and UL3 leads the gene-level table.

Figure 4. Frozen-panel coding mappings preserve overlapping annotations. The 257-guide panel produced 271 guide-to-CDS mappings; 250 guides mapped to at least one CDS. Values describe annotation coverage, not editing outcomes.

Figure 5. Configured panels show exact-target sequence barriers. All four three-guide panels required three distinct substitutions to remove every exact guide/PAM target. Held-out values are marginal per-guide observations. Barriers are not evolutionary probabilities.

Figure 6. Host-search rankings agree more closely than composite rankings. Cas-OFFinder and CRISPRitz were compared on the same 257 guides. ViralSafeTarget ranks include viral and annotation features and therefore address a broader prioritization axis; divergence is not evidence of superiority.

Additional files

- **Additional file 1:** Supplementary Figure S1 (PDF). Leave-one-component-out rank sensitivity on the frozen deep panel.
- **Additional file 2:** Machine-readable source tables (ZIP). Tables underlying the reported counts, rankings, virtual analyses and benchmark figures.
- **Additional file 3:** ViralSafeTarget 0.10.0 source archive (ZIP) with SHA-256 checksum, frozen from commit `de7868a83d3d1e30729323e53a735615d43fc231`.